



Chemical Analysis of Water

Introduction:

Water is one of the abundantly available substances in nature. It **contributes about 75% of the matter of earth's crust**. It is distributed in nature in various forms such as spring water, river water, rain water etc. Among these, the **purest form of water is rain water**. Its chemical formula is H_2O . It exists in gaseous, liquid and solid states. It has the unique ability to support to every kind of life as well as to dissolve variety of substances. So, it is regarded as **universal solvent**. Thus, it may contain variety of substances either in the form solution or suspensions which can distort the scientific analytical data. Most of these contaminants have the potential to affect the not only the number of scientific applications but also create a variety of problems in living as well as non-living systems *i.e.*, it affects the quality of water. So, to decide quality of water, *water analysis* is essential parameter of water analysis are commonly expressed in milligram per liter, part per million (ppm) etc. where **ppm means the number of parts of substance per million parts of water**. This unit is also called milligram per liter (mg/liter) or grams per cubic meter. Water analysis includes physical analysis and chemical analysis.



Chemical Analysis

Chemical analysis of water includes estimation of various chemical parameters of water. Here only total solids, (dissolved solids, suspended solids), hardness, salinity, alkalinity, acidity, sulphates, nitrates, dissolved oxygen, chemical oxygen demand, biological oxygen demand are discussed.

1) Total Solids:

Total solids mean both the dissolved and suspended *solids*. Commonly potable water contains dissolved minerals while sewage contains large amount of suspended or undissolved matter.

Suspended Solids (S.S.):

The small solid particles which remain in suspension in water as a colloid or due to motion of water called *suspended solids*. It is used as one indicator of water quality.

Method: 500 cm³ of sample of water is filtered through dried and weighed Gooch crucible (W₁). Suspended material retains in the crucible. It is washed with distilled water, so as to remove Cl⁻ ions. Then crucible is dried, weighed by taking usual precautions (W₂). The suspended solids are calculated by using the equation

$$\text{Wt. of suspended solids} = \frac{(W_2 - W_1) \times 10^6}{\text{Volume of sample taken}} = \frac{(W_2 - W_1) \times 10^6}{500} \text{ ppm}$$

Filtrate is collected along with washings to determine the dissolved solids.

Dissolved solids (DS) or Total dissolved solids (TDS):

Soluble salts of Na, K, Ca, Mg, Fe and Mn in the form of bicarbonates, carbonates, chlorides, nitrates, sulphates and phosphates etc. are present in natural water and increases the density, affects osmoregulation of organism, solubility of gases and hence it affects use of water for drinking, irrigation and industrial purposes.

Method: Filter 500 ml sample of water through Gooch crucible to make it free from suspended matter. Collect the filtrate and washings in a beaker and concentrate to about 50 ml. Transfer it to weighed evaporating dish (W_1). Evaporate on hot water bath to dryness. Note the weight of evaporating dish with residue by taking usual precautions (W_2). The TDS or DS is calculated by using the equation

$$\text{TDS or DS} = \frac{(W_2 - W_1) \times 10^6}{\text{Volume of sample taken}} = \frac{(W_2 - W_1) \times 10^6}{500} \text{ ppm}$$

Thus, Total solid = (Suspended solid + Dissolved solid) ppm.



2) Hardness:

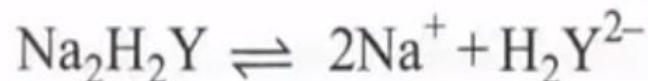
Water which does not give good lather with soap called *hard water*. Hardness is due to presence of salts like bicarbonates, chlorides and sulphates of Ca and Mg in it. Hardness is divided into temporary and permanent hardness where *temporary hardness* arises due to dissolved bicarbonates of Ca and Mg in water while *permanent hardness* is due to presence of dissolved sulphates and chlorides of Ca and Mg in water. Temporary hardness is known as *carbonate hardness* while permanent hardness is known as *non-carbonate hardness*. The total hardness is the sum of temporary and permanent hardness. Total hardness of water is expressed in terms of CaCO_3 ppm generally. Temporary hardness is removed by boiling the water, while to remove permanent hardness special treatment is essential.

Method: There are calculations, titrations, soap and complexometric methods of determination of hardness of water. Out of these *complexometric method* is more reliable. Hence it is discussed here.

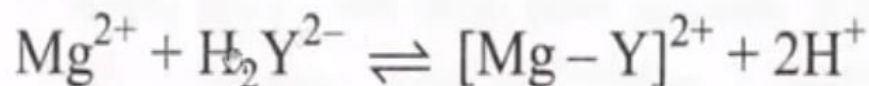
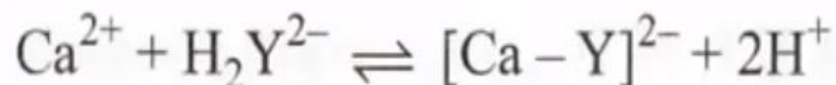
Hardness of water can be determined by titrating known quantity of hard water sample, against standard solution of EDTA by using Eriochrome Black T indicator, at pH 10.



EDTA is available as its disodium salt and it is represented as $\text{Na}_2\text{H}_2\text{Y}$ which ionizes as,



Resultant H_2Y^{2-} reacts with Ca^{2+} and Mg^{2+} ions and forms complexes as,



Here end point is wine red to sky blue, From the burette reading the hardness is calculated either as CaCO_3 in ppm or as mg equivalent per dm^3 . It is a total hardness of water.



3) Salinity:

Conceptually salinity of water may be defined as, it is the quantity of dissolved salts in water. It is also called as *saltiness*. It is measured in generally ppt or %. It varies from place to place in the ocean but relative proportions of the most dissolved major constituents generally remain common. The average ocean salinity is about 35 ppt; rivers and lakes can have wide range of salinities from 0.01ppt to 0.5 ppt. Dead Sea has salinity greater than 200 ppt. Here note that all-natural waters contain some dissolved salts of Na, Mg, Ca. Most common is NaCl constituent about 91%. It is the main constituent of sea water.

The level of salt in water affects its suitability for irrigation, stock, domestic uses etc. **There are two methods of determining the salt content of water** i.e., electrical conductivity (E.C.) and total dissolved solids (TDS). EC measurement can be done by passing an electric current between two metal plates or electrodes in the water sample while TDS is a measure of the relative weight of dissolved materials in water sample.

Here note that the term salinity is useful to oceanographers.



4) Alkalinity:

It is the acid neutralising capacity i.e., ANC of water. It arises due to presence of HCO_3^- , SiO_3^{2-} , HSiO_3^- , CO_3^{2-} . It is also due to humates. (Humate is salt of weak organic acid). Alkalinity due to HCO_3^- is called bicarbonate alkalinity; similarly carbonate alkalinity is due to CO_3^{2-} , hydrate alkalinity is due to OH^- .

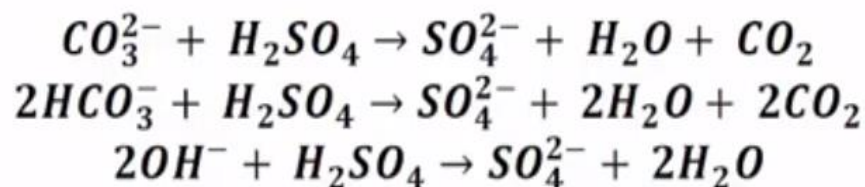
$$\text{ANC} = \text{Total alkalinity} = H_T$$

$$H_T = [\text{OH}^-] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}] - [\text{H}_3\text{O}^+]$$

In natural water species of B, Si, P, S etc. contribute in ANC but their contribution is too small.

Hydrate and carbonate alkalinity is also called phenolphthalein alkalinity as it is determined by it. Bicarbonate alkalinity is determined by methyl orange. So, it is called methyl orange alkalinity. It can be determined by titrating 50 ml sample against 0.02 N H_2SO_4 solution using phenolphthalein indicator. If necessary, filter the sample and then use. End point is pink to colourless. Then add few drops of methyl orange indicator and continue titration till end point is obtained (yellow to red). Total alkalinity, H_T is the sum of phenolphthalein alkalinity and methyl orange alkalinity.

Reactions:





Phenolphthalein alkalinity A_T can be calculated as,

$$A_T = \frac{N_1 V_1}{V_2} \times 1000 = \text{mg/dm}^3 \text{ in terms of CaCO}_3$$

Where N_1 = Normality of H_2SO_4 .

V_1 = Volume of H_2SO_4 when phenolphthalein indicator is used.

V_2 = Volume of sample taken.

$$H_T = \frac{N_1 V_3}{V_2} \times 1000 = \text{mg/dm}^3 \text{ in terms of CaCO}_3$$

Where V_3 = Total volume (phenolphthalein end point + Methyl orange end point).

Alkalinity may be predicted as.

- (a) If fraction end point for phenolphthalein indicator is zero, then A_T = Alkalinity may be due to bicarbonate ion only.
- (b) If there is no further titration, for methyl orange end point, after phenolphthalein end point, then alkalinity is due to Hydroxide ion.
- (c) If phenolphthalein end point is the half the total titration, then alkalinity is due to carbonate ion.
- (d) If phenolphthalein end point is greater than half the total titration, alkalinity is due to both carbonate and hydroxide ions.



(e) If phenolphthalein end point is less than half the total titration, alkalinity is due to carbonate and bicarbonate ions.

Alkalinity measurements of water are useful to decide-

- (a) buffering capacity of waste water and sludges.
- (b) suitability of water for irrigation.
- (c) the nature of salts presents in water.
- (d) the way for treatments of waste and natural water.
- (e) It is also useful to calculate Lange Lier saturation index.

Here, note that soft water obtained by orthophosphate treatment does not obey these rules. Instead of H_2SO_4 only strong acid like HCl can be used.



5) Acidity:

Acidity is the power of water to neutralise hydroxide (OH^-) ions. It is expressed in terms of calcium carbonate (CaCO_3). Acidity arises due to pickling liquors, acid mine drainage, industrial effluents, humic acid.

Acidity of water can be decided by titrating water sample with standard solution of NaOH. Methyl orange end point (pH 4.5) measures the acidity due to minerals and CO_2 while phenolphthalein end point (to reach pH 8.3) determines the total acidity.

Mineral acidity:

Here known quantity of water sample free from chlorine is titrated against 0.02 N NaOH using **methyl orange indicator**.

Total acidity at room temperature: Here water sample is titrated against 0.02 N NaOH using **phenolphthalein indicator**.

Total acidity at boiling temperature: To water sample add **phenolphthalein**, boil for 2 to 3 minutes and titrate with 0.02 N NaOH.

$$\text{Acidity as } \text{CaCO}_3 = \frac{\text{cm}^3 \text{ of } 0.02 \text{ N NaOH} \times 1 \times 1000}{\text{cm}^3 \text{ of water sample taken for titration}}$$



Note that **methyl orange acidity indicates mineral acidity**. If mineral acidity is absent then total acidity is CO_2 (Carbon dioxide) acidity of sample. **Phenolphthalein acidity is the total acidity of the sample water.**



6) Sulphate:

Natural water generally contains sulphate. Mine drainage wastes also contain high quantity of sulphate due to oxidation of pyrites. If potable water contains Na_2SO_4 and MgSO_4 beyond the prescribed limit it may cause cathartic action. It may be estimated gravimetrically, colorimetrically, turbidimetrically, titrimetrically. Here gravimetric method of sulphate estimation is discussed.

Gravimetric method: In this method sulphate is precipitated as BaSO_4 by the BaCl_2 solution (precipitating agent) in acidic (HCl) medium. Precipitation is carried out near the boiling temperature. Then the precipitate is filtered and washed with hot water to get free from Cl^- (Chloride) ions after digestion. Lastly this precipitate is ignited and weighed by taking usual precautions.



7) Nitrate:

It occurs in ground and surface water, waste water treatment plants, natural water distribution systems etc. It is the end products of decomposition of organic matter. Wastes from fertiliser manufacturing plants introduce nitrate in water system. It is also produced by aerobic stabilisation of organic nitrogen. *Its maximum admissible concentration (MAC) is 50 mg/litre.* If its concentration in water supply exceeds this limit then it has hazardous effect to health; dangerous to infants (below the age of six months), *causes a child disease methaemoglobinemia* etc. So, it becomes essential to monitor nitrate regularly in water supply.

There are several ways to estimate NO_3^- . One of the simple methods is discussed here. *i.e.*, In water sample phenol-di-sulphonic acid and KOH is added; the colour so formed is used to determine the concentration of NO_3^- in ppm. by comparing with standard colours. In spectrophotometric method the concentration of nitrate must be below 5 ppm or made so by diluting water sample.



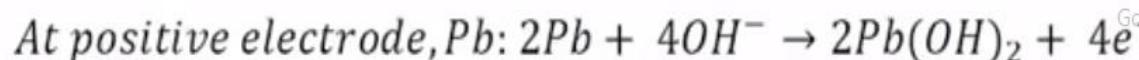
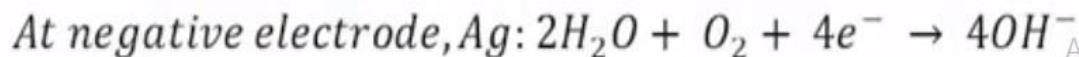
8) Dissolved Oxygen (DO):

It is understood that dissolved oxygen in water is the index of physical and biological processes going on in water. Water free from pollutants get saturated with oxygen (O_2). There are two ways by which Oxygen(O_2) get dissolved in water. 1) Diffusion/absorption from air 2) Due to photosynthesis in water by vegetation. Dissolved oxygen (DO) may be defined as,

'The amount of oxygen dissolved in milligram per dm^3 of water at fixed temperature is known as *dissolved oxygen*.' Oxygen dissolves in water slightly and is important for aquatic life. 4 to 6 mg per dm^3 of DO is the optimum value for good water quality, which ensures healthy aquatic life. Lower values of DO indicate water pollution. e.g., Within 4 to 5 ppm DO, fish becomes rare. Further decrease in DO causes migration of aquatic fauna and plant or death.

There are different methods used to decide DO. Here membrane electrode method is discussed. It consists of *Mackereth oxygen cell*. In this cell Ag and Pb electrodes are dipped in saturated $KHCO_3$ solution. These are separated by using polythene membrane (0.06 mm thick) from solution of sample test. Then galvanic cell is plugged of pH meter. It gives direct

DO reading in mg/dm^3 . Here readings are recorded for sample solution, standard and blank.





9) Chemical Oxygen Demand (COD):

It is the index of total organic content of water oxygen demanding substances in water. **It may be defined as the total amount of oxygen required for oxidation of substances is known as chemical oxygen demand.** It is based on the assumption that all organic matter can be oxidised by strong oxidising agents in acidic media. A variety of oxidising agents are used, but potassium dichromate is the best for this purpose. **COD is more scientific term than the term BOD.** Further it is reproducible. **It is expressed in ppm.**

Determination of COD:

The known quantity potassium dichromate ($K_2Cr_2O_7$) is added to the measured amount of sample. Mixture is boiled with concentrated sulphuric acid (H_2SO_4). Then excess of dichromate ions $Cr_2O_7^{2-}$ determined by titrating with standard ferrous ammonium sulphate solution (back titration). Indicator used is ferroin. As per need silver sulphate (Ag_2SO_4), Mercuric sulphate ($HgSO_4$) etc. are added to the solution. Difference between back and blank titration gives amount of dichromate consumed in oxidation of organic and inorganic matter. Thus,

$$COD \text{ mg/dm}^3 = \frac{(B - T) \times N \times 1000 \times 8}{\text{Volume of sample taken in cm}^3}$$



Where B = Blank reading, T = Back reading, and N = Normality
COD is useful

- 1) in management and design of treatment plant.
- 2) to determine the efficiency of treatment plants.
- 3) to study industrial waste.

pH of water, nitrification process, toxic materials and type of micro-organism affect the BOD test but COD test is not affected by these factors.



10) Biochemical or Biological Oxygen Demand (BOD):

Now days we are facing burning problem of water pollution. BOD determines degree of water pollution. Hence study of BOD of water becomes prime important. BOD may be defined as; 'The amount of oxygen required for the biological oxidation of organic material in water, during five days under controlled conditions (293 K) is known as *biochemical oxygen demand*.'

This is empirical parameter for organic matter in water. To get BOD value one has to wait for 5 days. The values are not reproducible. It is expressed in milligram per dm^3 oxygen or in ppm. For potable water it must be less than 1 ppm.

The substances decomposed in the BOD test may be the chemicals readily attacked by oxygen (O_2), such as sulphides, sulphates, sulphites, readily oxidisable compounds, ferrous ions etc. and food used by micro-organisms. The food of micro-organism may be nitrates and phosphates which, is found in domestic sewage, waste from animal feed lots, packing plants, farm wastes containing excess fertilisers etc. Increase in micro-organisms, increases value of BOD. High value of BOD is deleterious to aquatic lives.

The determination of BOD is useful to determine,

- 1) degree of pollution of water.
- 2) efficiency of waste water treatment plants.
- 3) self-purification capacity of water body.



- 4) possible control measures of pollution.
- 5) pollution load of waste water treatment on environment.
- 6) kind of organisms present in water.

Determination of BOD:

To determine BOD, BOD incubator, BOD bottles and allylthiourea solutions are essential. (To prepare Allylthiourea solution 500 mg allylthiourea dissolved in distilled water and diluted to 1 dm³.)

Method:

Take known quantity of sample. As per need adjust pH of sample to 7.0 by using one normal NaOH or one normal H₂SO₄ solution. To avoid loss of oxygen during incubation dilute the sample as per need. Now fill the two sets of BOD bottles. Add 1 cm³ allylthiourea solution to each bottle and stopper by taking usual precautions. Determine DO in one set immediately by using membrane electrode or any other method (DO). Incubate the other set in BOD incubator. After 5 days measure the DO (D₅).

$$BOD \text{ in } mg/dm^3 = (D - D_5 \times \text{Dilution factor}).$$

Here note that BOD is indicator and not pollutant.



Thanks